

Unit 1: Ratios and Proportions

Topic 1: Synthesis & Logic — Advanced Level 🍷🍷🍷

Name: _____

Date: _____

- Q1.** The ratio of x to y is 3:4. If x increases by 20%, what is the new ratio of x to y ?
- (A) 9:10
(B) 18:20
(C) 3.6:4
(D) 4:3
- Q2.** In a school, the ratio of students to teachers is 25:1. If there are 1200 students and 5 teachers leave, what is the new ratio?
- (A) 30:1
(B) 28:1
(C) 24:1
(D) 25:1
- Q3.** A gear with 12 teeth is meshed with a gear of 30 teeth. If the small gear makes 10 rotations, how many rotations does the large gear make?
- (A) 25
(B) 4
(C) 6
(D) 15
- Q4.** The ratio of the areas of two squares is 4:9. What is the ratio of their perimeters?
- (A) 4:9
(B) 16:81
(C) 2:3
(D) 8:18
- Q5.** In a bag, the ratio of red:blue:green marbles is 2:3:5. If there are 40 green marbles, how many marbles are NOT blue?
- (A) 24
(B) 56
(C) 80
(D) 48
- Q6.** Which coordinate point (x, y) represents a ratio where y is 150% of x ?
- (A) (2, 3)
(B) (3, 2)
(C) (1, 1.5)
(D) Both A and C
- Q7.** A is proportional to B . When $A = 12$, $B = 15$. Find A when $B = 40$.
- (A) 32
(B) 24
(C) 30
(D) 36
- Q8. [FRQ]** The ratio of the ages of two brothers is 3:5. In 4 years, the ratio will be 2:3. How old are they now?
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- Q9. [FRQ]** Mixture X is 2 parts juice and 3 parts water. Mixture Y is 3 parts juice and 7 parts water. If you mix 500ml of X and 500ml of Y, what is the final juice:water ratio?
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- Q10. [FRQ]** Solve: $12 : x = x : 27$. Find the positive value of x and explain its geometric meaning (Mean Proportional).
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💡 Chef's Tips (Advanced Level)

- **Inverse Ratios:** In gear problems, the number of rotations is inversely proportional to the number of teeth.
- **Area vs Length:** Ratio of lengths is the square root of the ratio of areas.
- **Age Problems:** Remember that years pass for everyone equally! Add the years to both sides of the ratio.